

Artificial Intelligence: Search, Reasoning & Learning

Models, formulas & decision rules. Read the full notes for derivations and worked examples.

01 / Match the frontier to the objective

BFS minimizes depth; UCS minimizes accumulated nonnegative path cost. Equal positive step costs align them. Test a UCS goal when removed from the priority queue, and account for duplicate states.

02 / A* needs a justified heuristic

Use $f=g+h$. Admissibility is a lower bound; consistency also constrains every edge. Graph search with inconsistent estimates may need reopening. Beam truncation sacrifices ordinary A* guarantees.

03 / Optimization is budgeted exploration

Hill climbing can stop locally. Annealing accepts a worsening cost Δ with $\exp(-\Delta/T)$. Genetic search needs valid encodings and diversity. A finite stochastic run is evidence of a found solution, not global optimality.

04 / Game backups

MAX takes maxima, MIN minima, chance nodes probability-weighted means. Alpha-beta prunes when alpha reaches beta for the same minimax tree. Heuristic cutoffs replace exact terminal guarantees with approximations.

05 / Prevent leakage

Fit preprocessing and model choices inside training and validation data. Keep the final test untouched. Use groups or time order when independent random splitting would misrepresent deployment.

06 / Evaluate the relevant errors

$\text{Precision} = \text{TP}/(\text{TP} + \text{FP})$, $\text{recall} = \text{TP}/(\text{TP} + \text{FN})$, $\text{F1} = 2\text{TP}/(2\text{TP} + \text{FP} + \text{FN})$. Name the positive class, handle empty denominators and weight errors when their consequences differ.

07 / Tree splits and pruning

Information gain is parent entropy minus size-weighted child entropy. Unique identifiers can look excellent in training. Validate numeric thresholds, depth and pruning; gain ratio is a mitigation, not a guarantee.

08 / Backpropagation bookkeeping

Save a forward pass, compute all gradients with those parameter values, then update. For sigmoid squared loss, output $\text{delta} = (a - t)a(1 - a)$. Hidden deltas sum weighted downstream sensitivities.

09 / Regularize deliberately

Learning rate affects stability; zero gradient need not indicate a minimum. Choose early stopping on validation. Inverted dropout scales retained training activations by $1/q$ and uses no dropout at inference.

10 / Margins and committees

Soft SVM balances norm against C-weighted slack; kernels must be valid inner products. Bagging reduces variance through resampling; boosting adapts emphasis; correlated errors limit the benefit of voting.

11 / Probability bookkeeping

Joint probabilities multiply local Bayesian-network factors. Sum hidden variables before normalizing. Conditioning differs from causation, and absence of a direct edge does not imply unconditional independence.

12 / Clustering choices change the answer

K-means pairs arithmetic means with squared Euclidean error. Single linkage can chain; complete linkage tests the farthest cross pair. Internal quality scores and cluster counts need interpretation and stability checks.